

JORDAN NEWMAN

Senior Software Engineer | Systems, Backend, Security, Infrastructure

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Selected public repositories are available. The most significant engineering repositories are private; access is available upon request.

SUMMARY

Senior software engineer specializing in systems software, distributed infrastructure, networking, security, and performance engineering. Built production systems spanning high-performance HTTP/CDN serving, FreeBSD kernel modules, observability, deployment automation, and enterprise integrations. Combines deep C and operating-system expertise with modern cloud delivery, developer tooling, and agentic AI systems.

TECHNICAL SKILLS

- **Languages:** C, C++, C#, Go, Python, Rust, TypeScript, JavaScript, Node.js, Shell, Objective-C, PHP, Perl, Visual Basic
- **Systems:** FreeBSD, Linux, kernel modules, system calls, operating systems, kqueue, sendfile(), GDB, TCP/IP, non-blocking I/O
- **Networking/Security:** HTTP, DNS, CDN architecture, socket programming, DDoS detection/mitigation, packet capture, libpcap, IPFW, authentication, telemetry
- **Cloud/DevOps:** Azure DevOps, Azure Pipelines, GitHub Actions, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation
- **AI/Developer Tooling:** LLM APIs and routing, MCP, agentic workflows, CLI tools, GitHub automation, auditable task execution
- **Databases/Web:** MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Built and operated infrastructure spanning approximately 3,000-4,000 servers, 10 CDN locations, and more than 65 Gbps of peak traffic.
- Designed the custom HTTP/CDN server and supporting software platform used for high-traffic production workloads.
- Built non-blocking event-driven systems around kqueue, zero-copy sendfile(), kernel-aware I/O, and dedicated disk I/O workers.
- Developed production FreeBSD kernel modules, DDoS mitigation systems, packet analyzers, DNS monitoring, authentication, and telemetry software.
- Built Nexus, a local-first agentic control plane with LLM routing, MCP integrations, declarative workflows, permissioned tools, and auditable task state.

SELECTED SYSTEMS PROJECTS

- **PrimeHTTPD:** Built a non-blocking C/FreeBSD HTTP/CDN server using kqueue, sendfile(), persistent connections, caching, and specialized I/O workers.
- **VirtualDir:** Built a FreeBSD kernel module that transparently remapped filesystem paths from runtime configuration; included the vdcli management tool.
- **PrimeDump:** Built a libpcap-based real-time traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 with ncurses views and IPFW integration.
- **PrimeDNSTop:** Built a DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source and domain activity.
- **Nexus:** Building a local-first agentic control plane with LLM routing, MCP integrations, declarative workflows, permissioned tools, and auditable task state.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist, APIs and Integrations

Remote / Tampa Bay, FL | June 2022 - Present

- Serve on the Architecture Team, contributing to technical decisions, engineering standards, shared libraries, platform direction, and documentation.
- Support about 12 teams with reusable Azure Pipelines, templates, deployment gates, and self-hosted runners that reduce manual work and enable quicker releases.
- Implement and improve Docker, Kubernetes, Terraform, CI/CD, security-scanning, developer-tooling, and AI-assisted engineering workflows.

DDI Systems - Senior Development Specialist

Manalapan, NJ | October 2021 - July 2022

- Modernized legacy VB.NET components in C# and integrated Microsoft Office 365 SMTP OAuth for secure customer-facing workflows.

ISPrime LLC - CEO

Weehawken, NJ | March 2018 - December 2019

- Led datacenter modernization, operational restructuring, and improvements to internally developed monitoring, security, and infrastructure platforms.

DDI Systems - Computer Programmer

Manalapan, NJ | April 2017 - March 2018

- Built ERP integrations and automated order, catalog, image, SDS, SFTP, SellerCloud, Essendant, and MultiValue/Universe data workflows.

Too Much Media LLC - Computer Programmer

Morganville, NJ | January 2016 - April 2017

- Developed real-time browser chat integrated with ticketing systems and implemented OAuth 2.0 authorization for protected sites.

MFCXY, Inc. - Computer Programmer

Chicago, IL | December 2014 - September 2015

- Developed backend and Windows client features and middleware for cross-platform database migration.

AJPM, LLC - Programmer / Owner

New Jersey / Remote | November 2012 - December 2013

- Built automated purchasing, inventory, repair, resale, and financial workflow software.

ISPrime Inc. - CIO / Partner

Weehawken, NJ | January 2001 - December 2014

Infrastructure Leadership

- Architected and operated hosting and CDN infrastructure spanning approximately 3,000-4,000 servers, 10 locations, multiple datacenters, and more than 65 Gbps of peak traffic.
- Led major technical projects and as many as 10 employees, taking unclear problems through architecture, hands-on implementation, and production operation.

Systems Software Engineering

- Designed and built PrimeHTTPD, a high-performance, non-blocking, kqueue-based HTTP/CDN server in C for FreeBSD; deployed it across approximately 200 servers and supported more than 150,000 concurrent connections.
- Leveraged `sendfile()`, `SF_NODISKIO`, `TCP_NODELAY`, `TCP_NOPUSH`, `O_NONBLOCK`, `accept_filter_http`, persistent connections, and dedicated I/O workers to keep disk operations from blocking the networking event loop.
- Developed production FreeBSD kernel modules including VirtualDir, KeepClean, and StatCache.

Security and Networking

- Built DDoS defenses, packet and DNS analysis, authentication, deployment, and production health-check and monitoring software including sitecheck, slugd, and php-seclogd.

FatWallet.com - Systems Administrator

Wisconsin / Remote | December 2000 - June 2001

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime and created automated availability and health alerting.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.